POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5033B | Semester V | Theory | 4 Credits

Special Electrical Machines

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5033B as Special Electical Machines (corrected to Electrical) in Semester V for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Delhi’s Special Electromechanical Systems course and polytechnic special machine curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Machine designs, drive parameters, electronic commutation logic and control settings must be based on approved manufacturer data, certified equipment testing, current safety standards and competent professional review.

Verified source note: Verified sources support stepper motors (VR, PM, Hybrid), switched reluctance motors, permanent magnet motors (BLDC and PMSM), and linear induction motors. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Special Electrical Machines studies the principles, construction and control of advanced motors used in precision automation and modern drives. It develops the ability to analyze stepper motors, switched reluctance motors (SRM), permanent magnet brushless DC (BLDC) motors and permanent magnet synchronous motors (PMSM), evaluate their torque-speed characteristics and control strategies while respecting the technical and safety boundaries of specialized machine operation.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Electrical machines I & II, power electronics, control systems, engineering electromagnetics and basic electronics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the construction, types and working principles of stepper motors for precision positioning.
2. Explain the working principle, torque production and control strategies of switched reluctance motors (SRM).
3. Explain the construction, electronic commutation and performance characteristics of permanent magnet brushless DC (BLDC) motors.
4. Explain the working principles of permanent magnet synchronous motors (PMSM), linear induction motors and other special AC machines.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the construction and stepping principles of various types of stepper motors.	Understand
CO2	Explain the torque production and power controller circuits for switched reluctance motors.	Understand
CO3	Explain the operation and electronic commutation of permanent magnet brushless DC motors.	Understand
CO4	Explain the characteristics of permanent magnet synchronous motors and linear induction motors.	Understand

Official CO–PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Stepper Motors and Control	Constructional features; types (Variable Reluctance, Permanent Magnet, Hybrid); working principle; step angle; resolution; drive circuits (unipolar, bipolar); applications in robotics and computer peripherals.	Approx. 14 hours	CO1	Module 1
2	Switched Reluctance Motors (SRM)	Constructional features; working principle; torque production; power controller circuits (converter types); control strategies; torque-speed characteristics; advantages and limitations.	Approx. 16 hours	CO2	Module 2
3	Permanent Magnet Brushless DC (BLDC) Motors	Constructional features; permanent magnet materials; electronic commutation; Hall sensors; torque and EMF equations; torque-speed characteristics; microprocessor-based control; applications in EVs and appliances.	Approx. 16 hours	CO3	Module 3
4	PMSM and Other Special Machines	Permanent Magnet Synchronous Motors (PMSM) construction and working; phasor diagram; control strategies (Vector control); Linear Induction Motors (LIM) principle and applications; Hysteresis motor; Repulsion motor.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Stepper Motors and Control

Constructional features; types (Variable Reluctance, Permanent Magnet, Hybrid); working principle; step angle; resolution; drive circuits (unipolar, bipolar); applications in robotics and computer peripherals.

CO1 · Approx. 14 hours

Switched Reluctance Motors (SRM)

Constructional features; working principle; torque production; power controller circuits (converter types); control strategies; torque-speed characteristics; advantages and limitations.

CO2 · Approx. 16 hours

Permanent Magnet Brushless DC (BLDC) Motors

Constructional features; permanent magnet materials; electronic commutation; Hall sensors; torque and EMF equations; torque-speed characteristics; microprocessor-based control; applications in EVs and appliances.

CO3 · Approx. 16 hours

PMSM and Other Special Machines

Permanent Magnet Synchronous Motors (PMSM) construction and working; phasor diagram; control strategies (Vector control); Linear Induction Motors (LIM) principle and applications; Hysteresis motor; Repulsion motor.

CO4 · Approx. 14 hours

MODULE 1

Stepper Motors and Control

Constructional features; types (Variable Reluctance, Permanent Magnet, Hybrid); working principle; step angle; resolution; drive circuits (unipolar, bipolar); applications in robotics and computer peripherals.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

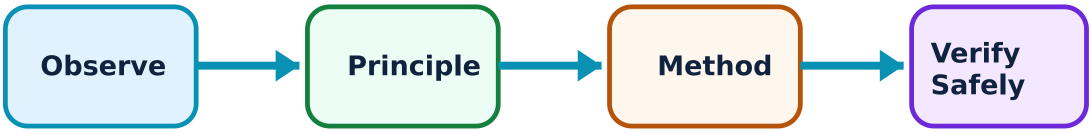
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Stepper Motors and Control: identify the system, state its principle, follow the correct method and verify the result safely.

1. Stepper motor types and principles−

Stepper motors convert electrical pulses into discrete mechanical movements. Variable Reluctance (VR) motors use a multi-tooth rotor. Permanent Magnet (PM) motors use a magnetized rotor. Hybrid motors combine both features for high resolution and torque. The step angle is the rotation per input pulse.

Study and application method

Calculate the step angle conceptually for a given number of stator and rotor teeth; compare unipolar and bipolar drive circuits.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the step angle conceptually for a given number of stator and rotor teeth; compare unipolar and bipolar drive circuits.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Stepper motors can lose steps if the load exceeds the pull-out torque. Do not operate motors at frequencies higher than the manufacturer's 'start-stop' rate.

2. Stepping modes and applications−

Stepping modes include full-step, half-step and micro-stepping (for smoother motion). Applications include 3D printers, CNC machines and camera focus mechanisms where precise incremental positioning is required.

Study and application method

Describe the excitation sequence for 'half-stepping' in a four-phase stepper motor; list three industrial applications of stepper motors.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the excitation sequence for 'half-stepping' in a four-phase stepper motor; list three industrial applications of stepper motors.

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Common mistake / precaution: Stepper motors generate significant heat during holding torque. Ensure authorised cooling and current-limiting measures are in place during operation.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Switched Reluctance Motors (SRM)

Constructional features; working principle; torque production; power controller circuits (converter types); control strategies; torque-speed characteristics; advantages and limitations.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

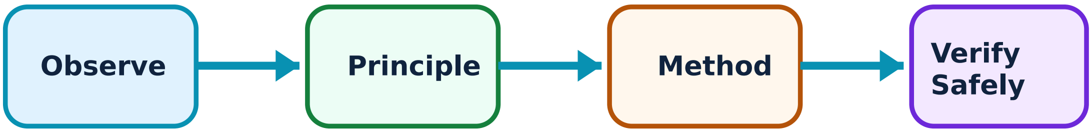
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Switched Reluctance Motors (SRM): identify the system, state its principle, follow the correct method and verify the result safely.

1. SRM construction and torque−

The Switched Reluctance Motor (SRM) has a simple, rugged construction with no windings or magnets on the rotor. Torque is produced by the tendency of the rotor to align with the energized stator phase to minimize magnetic reluctance.

Study and application method

Explain the principle of 'reluctance torque' production in an SRM; sketch the cross-section of a 6/4 SRM (6 stator poles, 4 rotor poles).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the principle of 'reluctance torque' production in an SRM; sketch the cross-section of a 6/4 SRM (6 stator poles, 4 rotor poles).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: SRMs produce significant torque ripple and acoustic noise. Use authorised control strategies and mechanical damping to mitigate these effects.

2. Converters and control−

SRMs require specialized power converters (e.g., asymmetric bridge) to energize the phases in sequence based on rotor position. Control strategies involve adjusting the turn-on and turn-off angles to optimize efficiency and torque.

Study and application method

Describe the operation of an asymmetric bridge converter for an SRM; explain the importance of 'rotor position sensing' in SRM drives.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the operation of an asymmetric bridge converter for an SRM; explain the importance of 'rotor position sensing' in SRM drives.

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Common mistake / precaution: Phase switching in SRMs involves high dV/dt and dI/dt. Ensure all converters have authorised snubber circuits and EMI shielding.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Permanent Magnet Brushless DC (BLDC) Motors

Constructional features; permanent magnet materials; electronic commutation; Hall sensors; torque and EMF equations; torque-speed characteristics; microprocessor-based control; applications in EVs and appliances.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

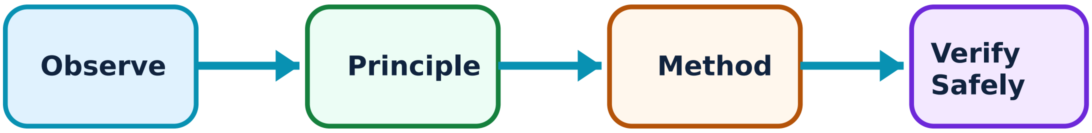
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Permanent Magnet Brushless DC (BLDC) Motors: identify the system, state its principle, follow the correct method and verify the result safely.

1. BLDC operation and commutation−

BLDC motors replace mechanical commutators with electronic ones. Hall effect sensors detect rotor position, and a controller switches the stator phases accordingly. This reduces maintenance, increases efficiency and allows for higher speeds.

Study and application method

Explain the process of 'electronic commutation' in a three-phase BLDC motor; describe the role of Hall sensors.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the process of 'electronic commutation' in a three-phase BLDC motor; describe the role of Hall sensors.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Incorrect commutation timing can lead to high current spikes and motor damage. Use only authorised controller firmware and sensor alignment procedures.

2. Characteristics and materials−

BLDC motors use high-energy magnets (e.g., Neodymium). They provide high power-to-weight ratios and flat torque-speed characteristics over a wide range, making them ideal for electric vehicles and battery-powered tools.

Study and application method

Sketch the torque-speed characteristic of a BLDC motor; list the advantages of BLDC motors over conventional brushed DC motors.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the torque-speed characteristic of a BLDC motor; list the advantages of BLDC motors over conventional brushed DC motors.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Permanent magnets can demagnetize at high temperatures. Do not exceed the authorised operating temperature limits of the motor.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

PMSM and Other Special Machines

Permanent Magnet Synchronous Motors (PMSM) construction and working; phasor diagram; control strategies (Vector control); Linear Induction Motors (LIM) principle and applications; Hysteresis motor; Repulsion motor.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

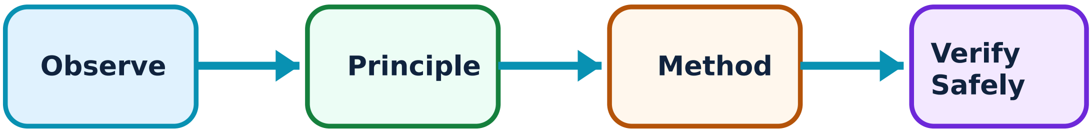
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for PMSM and Other Special Machines: identify the system, state its principle, follow the correct method and verify the result safely.

1. Permanent Magnet Synchronous Motors−

PMSMs are similar to BLDC but have a sinusoidal back-EMF and are driven by sinusoidal currents. They offer superior efficiency and smooth torque, often controlled using advanced techniques like Field Oriented Control (FOC) or Vector Control.

Study and application method

Compare PMSM and BLDC motors based on their back-EMF waveforms; explain the basic concept of 'Vector Control' in PMSM drives.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare PMSM and BLDC motors based on their back-EMF waveforms; explain the basic concept of 'Vector Control' in PMSM drives.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: PMSM control involves complex real-time calculations. Use authorised drive units and validated control algorithms for high-performance applications.

2. Linear motors and special AC machines−

Linear Induction Motors (LIM) produce motion in a straight line, used in maglev trains and conveyors. Hysteresis motors provide smooth, silent operation for timing devices. Repulsion motors offer high starting torque for specialized industrial tasks.

Study and application method

Describe the working principle of a Linear Induction Motor (LIM); explain why Hysteresis motors are preferred for silent operations.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the working principle of a Linear Induction Motor (LIM); explain why Hysteresis motors are preferred for silent operations.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Linear motors involve large magnetic air gaps and end-effects. All installations must meet authorised safety clearances and magnetic field exposure limits.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Stepper Motors and Control

Use the concept flow to revise observation, principle, method and verification.

Module 2: Switched Reluctance Motors (SRM)

Use the concept flow to revise observation, principle, method and verification.

Module 3: Permanent Magnet Brushless DC (BLDC) Motors

Use the concept flow to revise observation, principle, method and verification.

Module 4: PMSM and Other Special Machines

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare stepper motors and BLDC motors in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the cross-section of a permanent magnet rotor used in a PMSM.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the block diagram of a microprocessor-based control system for a BLDC motor.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the step angle conceptually for a hybrid stepper motor with 50 rotor teeth and 8 stator poles.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the type of special machine used in a computer hard disk drive or a modern washing machine.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Stepper Motors and Control

Explain Stepper motor types and principles and state one correct method, application or precaution.−

Stepper motors convert electrical pulses into discrete mechanical movements. Variable Reluctance (VR) motors use a multi-tooth rotor. Permanent Magnet (PM) motors use a magnetized rotor. Hybrid motors combine both features for high resolution and torque. The step angle is the rotation per input pulse.

Answer framework: Calculate the step angle conceptually for a given number of stator and rotor teeth; compare unipolar and bipolar drive circuits.

Important point: Stepper motors can lose steps if the load exceeds the pull-out torque. Do not operate motors at frequencies higher than the manufacturer's 'start-stop' rate.

Explain Stepping modes and applications and state one correct method, application or precaution.−

Stepping modes include full-step, half-step and micro-stepping (for smoother motion). Applications include 3D printers, CNC machines and camera focus mechanisms where precise incremental positioning is required.

Answer framework: Describe the excitation sequence for 'half-stepping' in a four-phase stepper motor; list three industrial applications of stepper motors.

Important point: Stepper motors generate significant heat during holding torque. Ensure authorised cooling and current-limiting measures are in place during operation.

Module 2 — Switched Reluctance Motors (SRM)

Explain SRM construction and torque and state one correct method, application or precaution.−

The Switched Reluctance Motor (SRM) has a simple, rugged construction with no windings or magnets on the rotor. Torque is produced by the tendency of the rotor to align with the energized stator phase to minimize magnetic reluctance.

Answer framework: Explain the principle of 'reluctance torque' production in an SRM; sketch the cross-section of a 6/4 SRM (6 stator poles, 4 rotor poles).

Important point: SRMs produce significant torque ripple and acoustic noise. Use authorised control strategies and mechanical damping to mitigate these effects.

Explain Converters and control and state one correct method, application or precaution.−

SRMs require specialized power converters (e.g., asymmetric bridge) to energize the phases in sequence based on rotor position. Control strategies involve adjusting the turn-on and turn-off angles to optimize efficiency and torque.

Answer framework: Describe the operation of an asymmetric bridge converter for an SRM; explain the importance of 'rotor position sensing' in SRM drives.

Important point: Phase switching in SRMs involves high dV/dt and dI/dt. Ensure all converters have authorised snubber circuits and EMI shielding.

Module 3 — Permanent Magnet Brushless DC (BLDC) Motors

Explain BLDC operation and commutation and state one correct method, application or precaution.−

BLDC motors replace mechanical commutators with electronic ones. Hall effect sensors detect rotor position, and a controller switches the stator phases accordingly. This reduces maintenance, increases efficiency and allows for higher speeds.

Answer framework: Explain the process of 'electronic commutation' in a three-phase BLDC motor; describe the role of Hall sensors.

Important point: Incorrect commutation timing can lead to high current spikes and motor damage. Use only authorised controller firmware and sensor alignment procedures.

Explain Characteristics and materials and state one correct method, application or precaution.−

BLDC motors use high-energy magnets (e.g., Neodymium). They provide high power-to-weight ratios and flat torque-speed characteristics over a wide range, making them ideal for electric vehicles and battery-powered tools.

Answer framework: Sketch the torque-speed characteristic of a BLDC motor; list the advantages of BLDC motors over conventional brushed DC motors.

Important point: Permanent magnets can demagnetize at high temperatures. Do not exceed the authorised operating temperature limits of the motor.

Module 4 — PMSM and Other Special Machines

Explain Permanent Magnet Synchronous Motors and state one correct method, application or precaution.−

PMSMs are similar to BLDC but have a sinusoidal back-EMF and are driven by sinusoidal currents. They offer superior efficiency and smooth torque, often controlled using advanced techniques like Field Oriented Control (FOC) or Vector Control.

Answer framework: Compare PMSM and BLDC motors based on their back-EMF waveforms; explain the basic concept of 'Vector Control' in PMSM drives.

Important point: PMSM control involves complex real-time calculations. Use authorised drive units and validated control algorithms for high-performance applications.

Explain Linear motors and special AC machines and state one correct method, application or precaution.−

Linear Induction Motors (LIM) produce motion in a straight line, used in maglev trains and conveyors. Hysteresis motors provide smooth, silent operation for timing devices. Repulsion motors offer high starting torque for specialized industrial tasks.

Answer framework: Describe the working principle of a Linear Induction Motor (LIM); explain why Hysteresis motors are preferred for silent operations.

Important point: Linear motors involve large magnetic air gaps and end-effects. All installations must meet authorised safety clearances and magnetic field exposure limits.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Stepper Motors and Control.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.

3. State one key definition from Module 2: Switched Reluctance Motors (SRM).
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Permanent Magnet Brushless DC (BLDC) Motors.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: PMSM and Other Special Machines.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Constructional features; types (Variable Reluctance, Permanent Magnet, Hybrid); working principle; step angle; resolution; drive circuits (unipolar, bipolar); applications in robotics and computer peripherals..
2. Develop a complete answer or practical plan for: Constructional features; working principle; torque production; power controller circuits (converter types); control strategies; torque-speed characteristics; advantages and limitations..
3. Develop a complete answer or practical plan for: Constructional features; permanent magnet materials; electronic commutation; Hall sensors; torque and EMF equations; torque-speed characteristics; microprocessor-based control; applications in EVs and appliances..
4. Develop a complete answer or practical plan for: Permanent Magnet Synchronous Motors (PMSM) construction and working; phasor diagram; control strategies (Vector control); Linear Induction Motors (LIM) principle and applications; Hysteresis motor; Repulsion motor..

LAST-MILE REVISION

Quick Revision

Module 1: Stepper Motors and Control

Constructional features; types (Variable Reluctance, Permanent Magnet, Hybrid); working principle; step angle; resolution; drive circuits (unipolar, bipolar); applications in robotics and computer peripherals.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Switched Reluctance Motors (SRM)

Constructional features; working principle; torque production; power controller circuits (converter types); control strategies; torque-speed characteristics; advantages and limitations.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Permanent Magnet Brushless DC (BLDC) Motors

Constructional features; permanent magnet materials; electronic commutation; Hall sensors; torque and EMF equations; torque-speed characteristics; microprocessor-based control; applications in EVs and appliances.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: PMSM and Other Special Machines

Permanent Magnet Synchronous Motors (PMSM) construction and working; phasor diagram; control strategies (Vector control); Linear Induction Motors (LIM) principle and applications; Hysteresis motor; Repulsion motor.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Stepper motor types and principles	Stepper motors convert electrical pulses into discrete mechanical movements.
Stepping modes and applications	Stepping modes include full-step, half-step and micro-stepping (for smoother motion).
SRM construction and torque	The Switched Reluctance Motor (SRM) has a simple, rugged construction with no windings or magnets on the rotor.
Converters and control	SRMs require specialized power converters (e.
BLDC operation and commutation	BLDC motors replace mechanical commutators with electronic ones.
Characteristics and materials	BLDC motors use high-energy magnets (e.
Permanent Magnet Synchronous Motors	PMSMs are similar to BLDC but have a sinusoidal back-EMF and are driven by sinusoidal currents.
Linear motors and special AC machines	Linear Induction Motors (LIM) produce motion in a straight line, used in maglev trains and conveyors.

LEARNING RESOURCES

References

Recommended special-machines references

1. T. J. E. Miller, Brushless Permanent Magnet and Reluctance Motor Drives.
2. K. Venkataratnam, Special Electrical Machines.
3. E. G. Janardhanan, Special Electrical Machines.
4. R. Krishnan, Electric Motor Drives: Modeling, Analysis and Control.

Approved public special-machines sources

- https://nptel.ac.in/courses/108102156
- https://nptel.ac.in/courses/108102372

These sources provide the verified study basis while official Course 5033B detail is unavailable; they do not replace official equipment manuals, authorised machine testing, certified designs or authorised industrial control protocols.